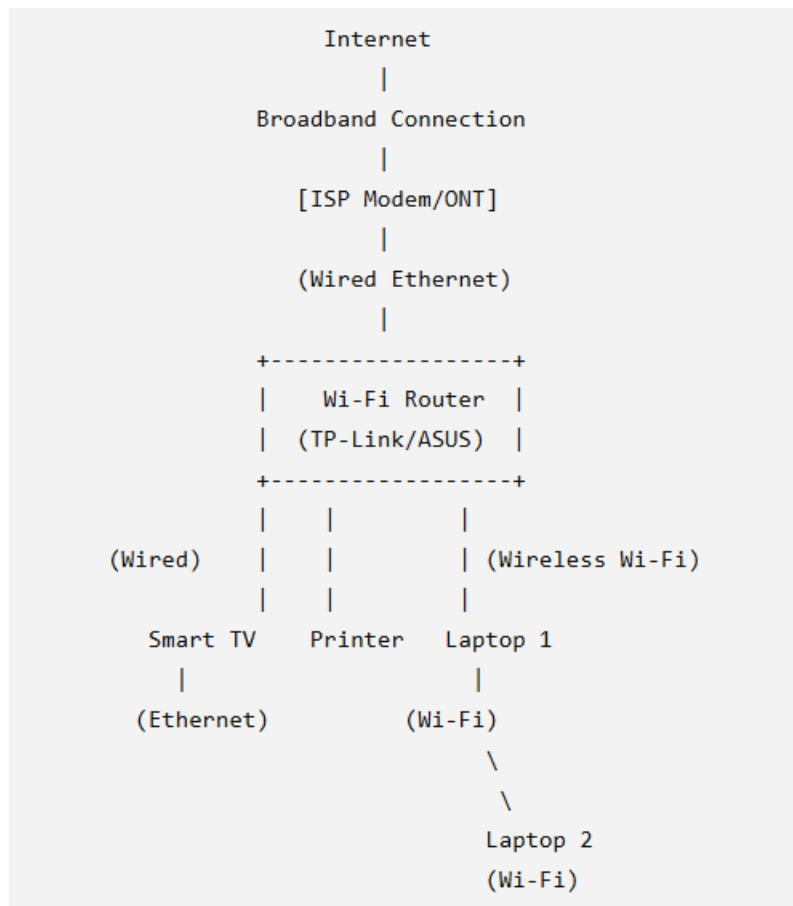


Build a SOHO Network

Task 1.

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).



ISP Modem → Wi-Fi Router: Wired (Ethernet)

Router → Smart TV: Wired (Ethernet)

Router → Printer: Wired (Ethernet)

Router → Laptop 1: Wireless (Wi-Fi)

Router → Laptop 2: Wireless (Wi-Fi)

Task 2.

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

Hardware Components Required

Component	Suggested Brand/Model
Broadband Fiber Connection	Airtel Xstream Fiber / JioFiber / BSNL FTTH
Wi-Fi Router	TP-Link Archer C6 (Wi-Fi 5) or ASUS RT-AX53U (Wi-Fi 6)
ISP Modem/ONT	Nokia/Genexis ONT (provided by ISP)
Laptop (2)	Dell Inspiron / HP 15s
Smart TV	Samsung 43-inch Smart TV
Printer	HP Smart Tank 580 or Epson EcoTank L3250
Cat6 Ethernet Cable (2–3)	D-Link / UGREEN
RJ45 Connectors (optional)	D-Link
5-Port Gigabit Switch (if more LAN ports needed)	TP-Link TL-SG105
Wi-Fi Range Extender (optional)	TP-Link RE205
Power Extension Board	Anchor / Havells

Task 3.

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

Step 1: Connect the Router

- Connect ISP modem/ONT to the router's WAN port.
- Connect your computer using Wi-Fi or LAN cable.

Step 2: Open Router Settings

- Open a web browser.
- Enter:
 - 192.168.1.1
- Login using the default username and password.

Step 3: Change Admin Password

- Navigate to Administration/System Tools.
- Change the default administrator password to a strong password.
- Save the settings.

Step 4: Configure Internet

- Select the connection type provided by your ISP (DHCP or PPPoE).
- Enter ISP username and password if required.

Step 5: Set Wi-Fi Name (SSID)

- Go to Wireless Settings.
- Set:
 - SSID: SOHO_Office
- Save.

Step 6: Enable WPA2 Security

- Select:
 - Security Mode: WPA2-PSK
 - Encryption: AES
- Set a strong Wi-Fi password.
- Example:

Tops@1234

- Save settings.

Step 7: Enable DHCP

- DHCP Server: Enabled
- DHCP Range:
- 192.168.1.100
- to

192.168.1.200

Step 8: Reboot Router

- Restart the router.
- Connect all devices to the new Wi-Fi.

Task 4.

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Device	MAC Address	IP Address	Type
Wi-Fi Router	00:1A:2B:3C:4D:5E	192.168.1.1	Gateway
Laptop 1	A4:5E:60:11:22:33	192.168.1.101	DHCP
Laptop 2	B8:27:EB:44:55:66	192.168.1.102	DHCP
Smart TV	C2:4D:66:77:88:99	192.168.1.20	Static
Printer	D8:9A:BC:10:20:30	192.168.1.10	Static

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 8.8.8.8

Task 5.

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

Cause 1: Wi-Fi is Disabled

Diagnosis

- Check whether the Wi-Fi icon is turned off.
- Verify Airplane Mode is disabled.

Solution

- Enable Wi-Fi.
- Reconnect to the correct SSID.
- Restart the laptop if necessary.

Cause 2: Incorrect Wi-Fi Password

Diagnosis

- Laptop shows "Authentication Failed" or repeatedly asks for the password.

Solution

- Forget the Wi-Fi network.
- Re-enter the correct WPA2 password.
- Ensure the correct SSID is selected.

Cause 3: IP Address Not Assigned

Diagnosis

Open Command Prompt and run:

```
ipconfig
```

If the IP address starts with 169.254.x.x, DHCP has failed.

Solution

Run:

```
ipconfig /release
```

ipconfig /renew

Restart the router if the issue persists.

Cause 4: Router Has No Internet

Diagnosis

- Check whether other devices can access the internet.
- Verify the WAN/Internet LED on the router.

Solution

- Restart the modem and router.
- Check ISP service status.
- Reconfigure PPPoE credentials if needed.

Cause 5: DNS Problem

Diagnosis

Run:

ping 8.8.8.8

If this works but websites do not open, DNS may be the issue.

Solution

Configure DNS manually:

- Preferred DNS: 8.8.8.8
- Alternate DNS: 1.1.1.1

Cause 6: Network Adapter Issue

Diagnosis

Open Device Manager and check for errors on the Wi-Fi adapter.

Solution

- Update or reinstall the Wi-Fi driver.
- Disable and re-enable the adapter.
- Restart the laptop.